Lab 3 ticket

Nutrients, LU, coded RREF — Friday, October 30

Lab 3 · Fri Oct 30 · ~20–30 min · Laptop + paper · Initials

Name: _____ **Language (circle):** Octave / Python

Oats/yogurt; protein/fiber. $A = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$. Mon want $(26, 10)$; Tue want $(31, 12)$.

Lab 2 system later: $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, $b = (6, 11, 14)$.

New syntax

I needed to…	Command / pattern
One entry M_{ij}	`E(k,k)` or `E[k,k]`
Last column	`M(:,end)` or `M[:,-1]`
`for` loop	(+ `end` / indent)
Print	(`disp`/`print`, `num2str` / f-string)
Augmented M; build E; left-multiply	

C3.1 Mon. Want 26 g protein, 10 g fiber. Servings $x =$ _____ **Exp:** $(4, 6)$

C3.2 Tue. Want 31 g protein, 12 g fiber. Servings $x' =$ _____ **Exp:** $(5, 7)$

C3.3 LU. $LU = A$? Y / N Mon: $Ly = b$ then $Ux = y$ gives same x? Y / N

C3.4 RREF code. Your $x =$ _____ `A\b` / `solve` same? Y / N **Exp:** $(1, 2, 3)$
One line: `A\b` is really _____

C3.5 $[A \mid I]$**.** $BA = I$? Y / N $Bb = x$? Y / N $B =$ product of E_i? Y / N
Why prefer solver / LU over forming A^{-1}? _____

C3.6 Powers. $A^3 x_0 =$ _____ **Exp:** $(6, 1)$

Stuck on: _____

Check-in initials: _____